

ogies, more collaboration and information sharing will help the community innovate faster and more inclusively, and identify specific risks, to mitigate them before putting a product into the world," Rossi emphasised.

One might wonder how these high-level principles translate into practical measures within IBM's AI systems. Rossi provided concrete examples: "IBM has developed practitioner-friendly bias mitigation approaches, proposed methods for understanding differences between AI models in an interpretable manner, studied maintenance of AI models from the robustness perspective, and created methods for understanding the activation space of neural networks for various trustworthy AI tasks."

She also mentioned that IBM has analysed adversarial vulnerabilities in AI models and proposed training approaches to mitigate such vulnerabilities. "We made significant updates to our AI explainability 360 toolkit to support time series and industrial use cases, and have developed application-specific frameworks for trustworthy AI," she added.

AI INNOVATION WITHIN ETHICAL BOUNDARIES

A common concern is whether strict ethical guidelines stifle innovation. Rossi sees ethics as an enabler rather than a hindrance. "AI can drive tremendous progress for business and society – but only if it's trusted," she stated.

She cited IBM's annual Global AI Adoption Index, noting that while 42% of enterprise-scale companies have deployed AI, 40% are still exploring or experimenting without deployment. "Ongoing challenges for AI adoption in enterprises remain, including hiring employees with the right skill sets, data complexity, and ethical concerns," Rossi said. "Companies must prioritise AI ethics and trustworthy AI to successfully deploy the technology and encourage further innovation."

Building AI systems that prioritise ethical considerations is no small feat. Rossi acknowledged the hurdles: "We see a large percentage of companies stuck in the experimentation and

exploration phase, underscoring a dramatic gap between hype around AI and its actual use."

She pointed out that challenges like the skills gap, data complexity, and AI trust and governance are significant barriers. "IBM's annual Global AI Adoption Index recently found that while around 85% of businesses agree that trust is key to unlocking AI potential, well under half are taking steps towards truly trustworthy AI, with only 27% focused on reducing bias," she noted.

To address these challenges, IBM launched watsonx, an enterprise-ready AI and data platform. "It accelerates the development of trusted AI and provides the visibility and governance needed to ensure that AI is used responsibly," Rossi explained.

INDIA'S ROLE IN SHAPING GLOBAL AI ETHICS

India is rapidly emerging as a major player in AI innovation, and Rossi believes the country has a significant role to play in shaping global AI ethics and governance.

"Given that the AI market in India is growing at a rapid pace, with some estimates suggesting it is growing at a CAGR of 25-35% and expected to reach 17 billion USD by 2027, AI ethics and governance will be key as the market continues to develop," she said.

She highlighted recent initiatives like the Global IndiaAI Summit 2024 conducted by the Ministry of Electronics and Information Technology (MeitY), which focused on advancing AI development in areas like compute capacity, foundational models, datasets, application development, future skills, startup financing, and safe AI.

With India's growing talent pool in AI and data science, education and training in AI ethics are paramount. Rossi mentioned that IBM researchers in India are focused on AI ethical challenges across IBM's three labs in the country: IBM Research India, IBM India Software Labs, and IBM Systems Development Labs.

"These labs are closely aligned to our strategy, and their pioneering work in AI,

Cloud, Cybersecurity, Sustainability, Automation is integrated into IBM products, solutions, and services," she said.

FUTURE OF AI ETHICS

Looking ahead, Rossi is optimistic but cautious about the evolution of AI ethics over the next decade. "Investing in AI ethics is crucial for long-term profitability, as ethical AI practices enhance brand reputation, build trust, and ensure compliance with evolving regulations," she asserted.

IBM is actively building a robust ecosystem to advance ethical, open innovation around AI. "We recently collaborated with Meta and more than 120 other open-source leaders to launch the AI Alliance, a group whose mission is to build and support open technology for AI and the open communities that will enable it to benefit all of us," Rossi shared.

As AI becomes increasingly interconnected and embedded in our lives, new ethical challenges will arise. Rossi highlighted the importance of focusing on trust in the era of powerful foundation models.

"In keeping with our focus on trustworthy AI, IBM is developing solutions for the next challenges such as robustness, uncertainty quantification, explainability, data drift, privacy, and concept drift in AI models," she said.

The TLDR version of my interview with IBM's Francesca Rossi constantly underscores a fundamental truth: ethical considerations in AI are not optional – they're essential for sustainable success. As Rossi aptly put it, "These considerations are not in opposition to profit but are rather essential for sustainable success."

In a world where AI's influence is only set to grow, Francesca Rossi's insights offer a roadmap for navigating the complex ethical landscape of AI. It's a journey that demands transparency, collaboration, and an unwavering commitment to building systems that not only advance technology but also uphold the values that define us as a society. But it's a collective effort that involves policymakers, educators, and industry leaders here in India and around the world. 